

University OF Engineering and Technology, Lahore
Computer Engineering Department

Course Name: PF	Course Code: CMPE-112L
Assignment Type: lab	Dated: 16 FEB 2026
Semester: 1st	Session: Spring 2026
Lab/Project/Assignment #: 1	CLOs to be covered: CLO1
Lab Title: input/Output, Data Types, Operators, Conditional Statements (If/Else)	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB NO 1

Lab Goals/Objectives:

- **Input/Output:** Giving your program a voice (printing) and a way to listen (taking user input).
- **Data Types:** Teaching your program the difference between text, integers, and decimals so it knows how to handle information.
- **Operators:** Giving your program the ability to do math and compare values.
- **Conditional Statements:** Giving your program a brain to make decisions based on the data it receives.

Equipment Required:

- Computer/Laptop
- Python 3.x
- VS Code / PyCharm / IDLE

Theory:

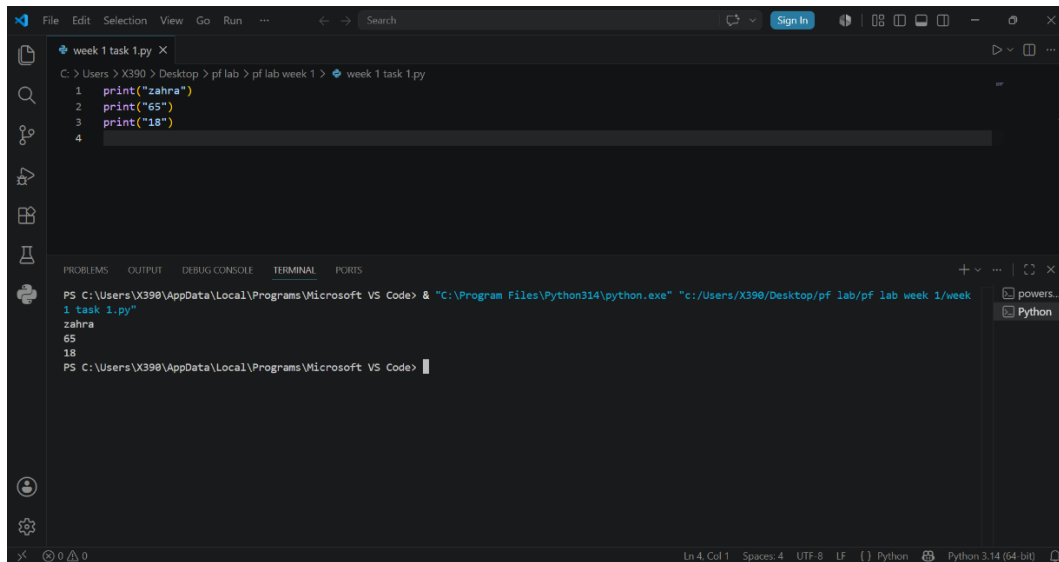
This lab explores foundational programming principles: using **Input/Output** (`input/print`) to communicate with users, managing **Data Types** (`int`, `float`, `str`) through type casting, and utilizing **Operators** to perform math or evaluate relationships. These concepts culminate in **Conditional Statements** (`if/elif/else`), which use indentation to introduce logical decision-making paths that selectively execute code based on whether conditions are true or false

Lab Work:

Task 1:

Write a Python program for a fitness club that takes a user's name, weight, and age. The program needs to process this data and use conditional logic to determine their membership eligibility

RESULT:



The screenshot shows the Microsoft VS Code interface. The editor window displays a file named 'week 1 task 1.py' with the following Python code:

```
1 print("zahra")
2 print("65")
3 print("18")
4
```

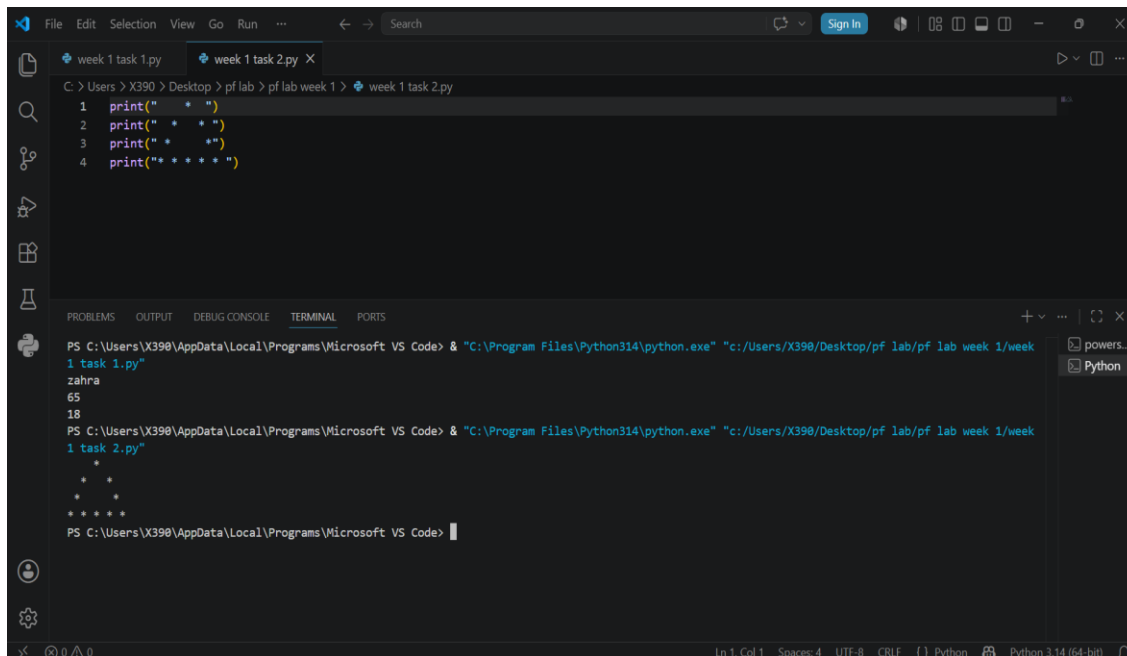
The terminal window at the bottom shows the command prompt output after running the script:

```
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 1/week 1 task 1.py"
zahra
65
18
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>
```

Task 2:

Write a Python program to practice basic console output manipulation and string formatting by generating a structured geometric shape using a single character.

RESULT:



The screenshot shows the Microsoft VS Code interface. The editor window displays a file named 'week 1 task 2.py' with the following Python code:

```
1 print(" * ")
2 print(" * * ")
3 print(" * * * ")
4 print(" * * * * ")

```

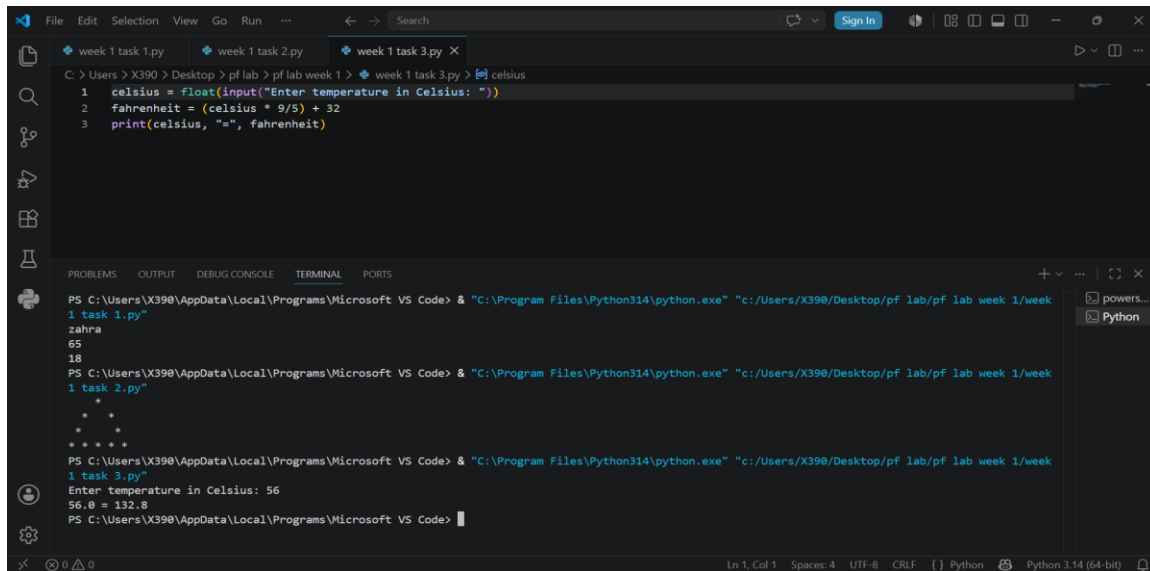
The terminal window at the bottom shows the command prompt output after running the script:

```
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 1/week 1 task 2.py"
 * 
 * * 
 * * * 
 * * * * 
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>
```

TASK NO 3:

Write a Python program that prompts the user for a temperature in Celsius, converts that value into Fahrenheit using standard mathematical operators, and outputs the result.

RESULT:



The screenshot shows the Visual Studio Code interface. The editor window displays a Python file named 'week 1 task 3.py' with the following code:

```
1 celsius = float(input("Enter temperature in Celsius: "))
2 fahrenheit = (celsius * 9/5) + 32
3 print(celsius, "=", fahrenheit)
```

The terminal window at the bottom shows the command prompt running the script. The output is as follows:

```
PS C:\Users\X390\Desktop> cd pf lab week 1 & python week 1 task 3.py
Enter temperature in Celsius: 56
56.0 = 132.8
```

TASK NO 4:

Write a Python program to practice handling user interaction by collecting multiple pieces of personal data and immediately displaying them back to the user.

RESULT:

```
1 Name = input("enter your name")
2 Age = input("enter your age")
3 Roll = input(" enter your roll no")
4 print(Name)
5 print(Age)
6 print(Roll)
```

```
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 1/week 1 task 3.py"
Enter temperature in Celsius: 56
56.0 = 132.8
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 1/week 1 task 4.py"
enter your namezahra
enter your age19
enter your roll no79
zahra
19
79
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>
```

TASK NO 5:

Write a Python program that accepts a data storage size in bytes from the user, converts that value into Megabytes (MB) and Gigabytes (GB) using binary size conversions, and displays the results.

RESULT:

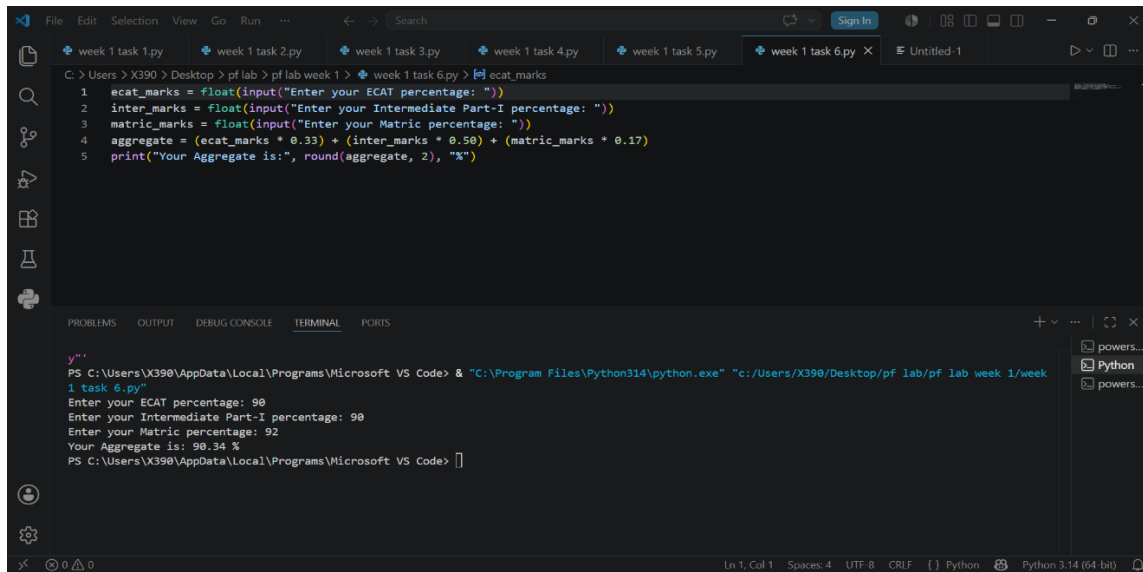
```
1 bytes = float(input("Enter the number of bytes: "))
2 megabytes = bytes / (1024 * 1024)
3 gigabytes = bytes / (1024 * 1024 * 1024)
4 print("Megabytes (MB):", megabytes)
5 print("Gigabytes (GB):", gigabytes)
```

```
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 1/week 1 task 5.py"
Enter the number of bytes: 79
Megabytes (MB): 7.534027099609375e-05
Gigabytes (GB): 7.35744833946228e-08
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>
```

TASK NO 6:

Write a Python program that calculates a student's final university admission aggregate score based on predefined percentage weights from three academic criteria: ECAT entry test results, Intermediate Part-I marks, and Matriculation marks.

RESULT:



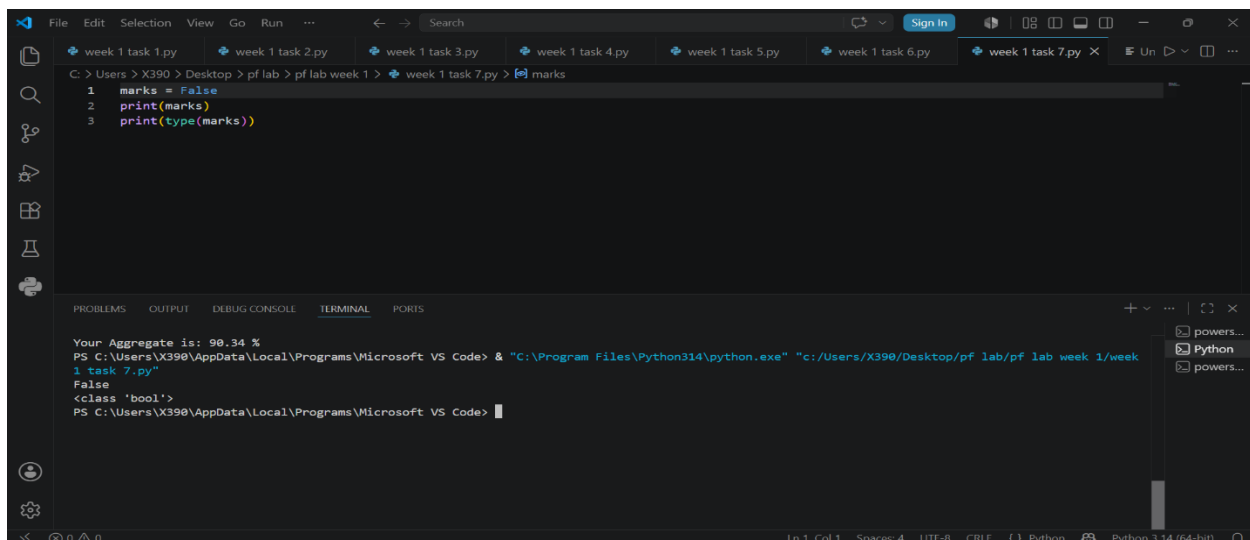
```
File Edit Selection View Go Run ... Search
week 1 task 1.py week 1 task 2.py week 1 task 3.py week 1 task 4.py week 1 task 5.py week 1 task 6.py X Untitled-1
C: > Users > X390 > Desktop > pf lab > pf lab week 1 > week 1 task 6.py > ecate_marks
1 ecate_marks = float(input("Enter your ECAT percentage: "))
2 inter_marks = float(input("Enter your Intermediate Part-I percentage: "))
3 matric_marks = float(input("Enter your Matric percentage: "))
4 aggregate = (ecate_marks * 0.33) + (inter_marks * 0.50) + (matric_marks * 0.17)
5 print("Your Aggregate is:", round(aggregate, 2), "%")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
y""
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 1/week 1 task 6.py"
Enter your ECAT percentage: 90
Enter your Intermediate Part-I percentage: 90
Enter your Matric percentage: 92
Your Aggregate is: 90.34 %
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> []
```

TASK NO 7:

Write a Python program to demonstrate variable assignment with logical data values and use built-in inspection functions to verify its underlying data type.

RESULT:



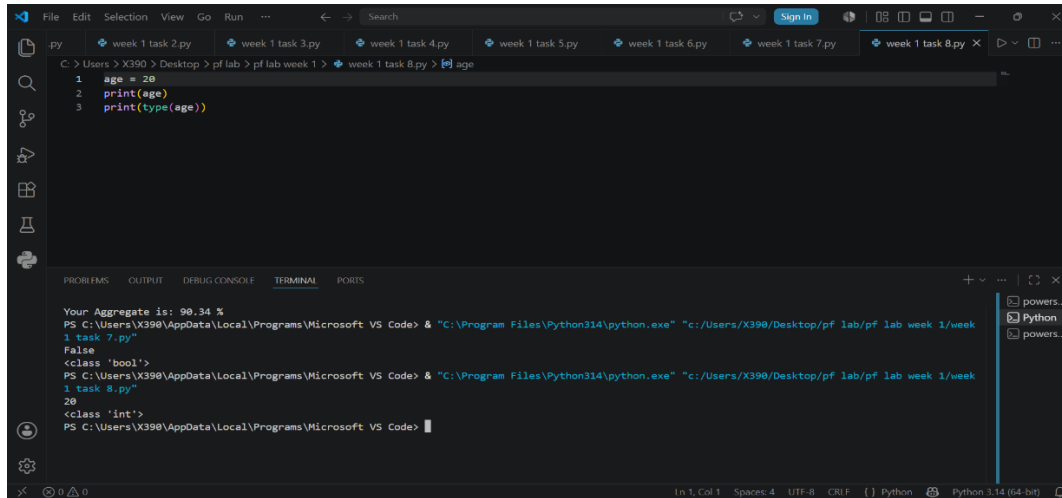
```
File Edit Selection View Go Run ... Search
week 1 task 1.py week 1 task 2.py week 1 task 3.py week 1 task 4.py week 1 task 5.py week 1 task 6.py week 1 task 7.py X Un
C: > Users > X390 > Desktop > pf lab > pf lab week 1 > week 1 task 7.py > marks
1 marks = False
2 print(marks)
3 print(type(marks))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Your Aggregate is: 90.34 %
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 1/week 1 task 7.py"
False
<class 'bool'>
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> []
```

TASK NO 8:

Write a Python program to demonstrate variable assignment with numeric data and use built-in inspection functions to verify its underlying data type class

RESULT:



The screenshot shows the Visual Studio Code editor with a Python file named 'week 1 task 8.py'. The code in the editor is:

```
1 age = 20
2 print(age)
3 print(type(age))
```

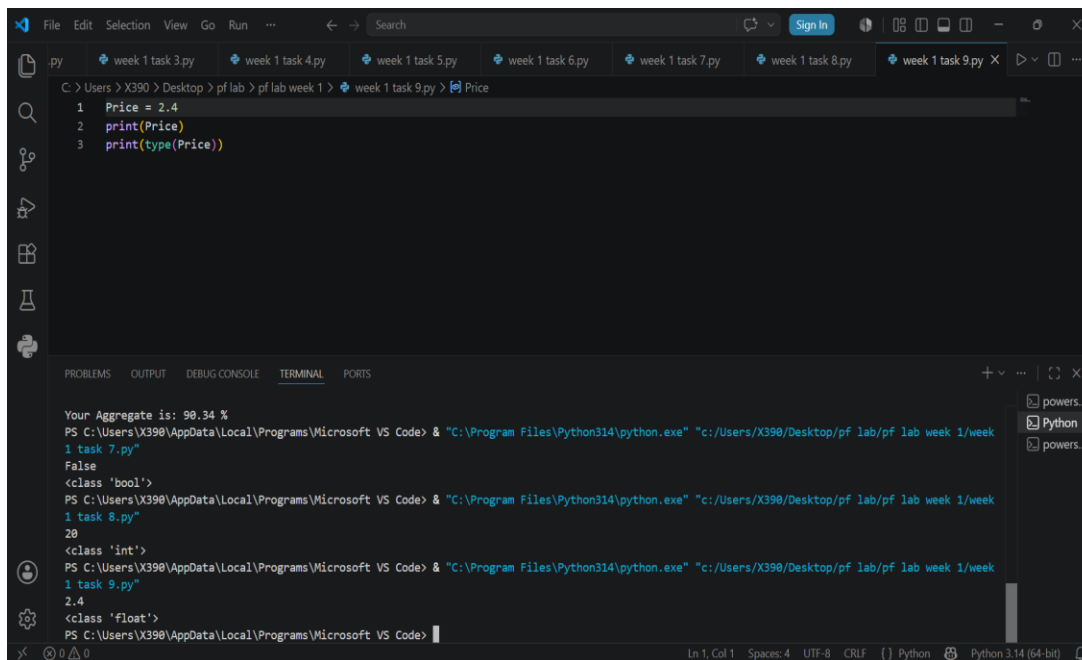
The terminal output at the bottom shows the execution of the script:

```
Your Aggregate is: 90.34 %
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 1/week 1 task 7.py"
False
<class 'bool'>
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 1/week 1 task 8.py"
20
<class 'int'>
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>
```

TASK NO 9:

Write a Python program to demonstrate variable assignment with decimal numeric data and use built-in inspection functions to verify its underlying data type class.

RESULT:



The screenshot shows the Visual Studio Code editor with a Python file named 'week 1 task 9.py'. The code in the editor is:

```
1 Price = 2.4
2 print(Price)
3 print(type(Price))
```

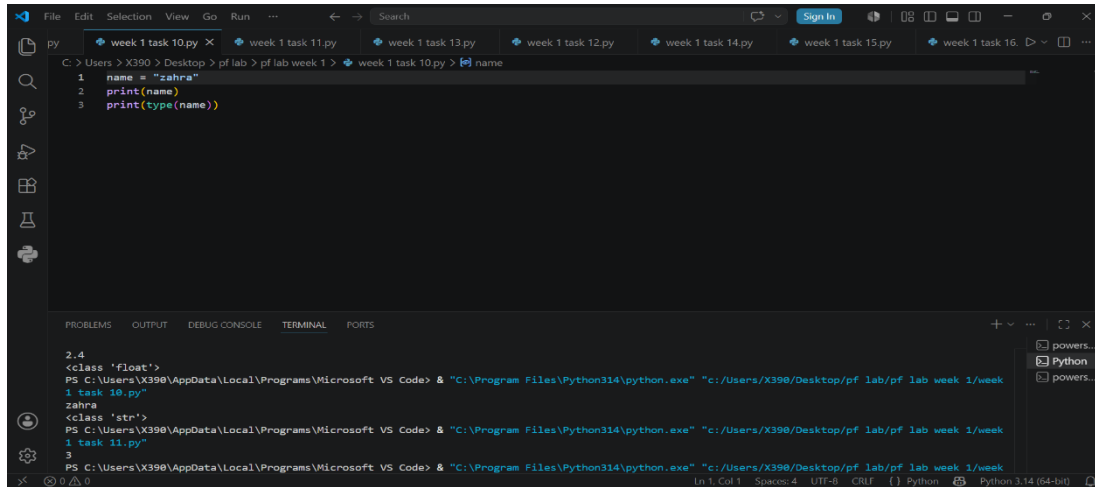
The terminal output at the bottom shows the execution of the script:

```
Your Aggregate is: 90.34 %
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 1/week 1 task 7.py"
False
<class 'bool'>
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 1/week 1 task 8.py"
20
<class 'int'>
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 1/week 1 task 9.py"
2.4
<class 'float'>
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>
```

TASK NO 10:

Write a Python program to demonstrate variable assignment with textual data and use built-in inspection functions to verify its underlying data type class

RESULT:



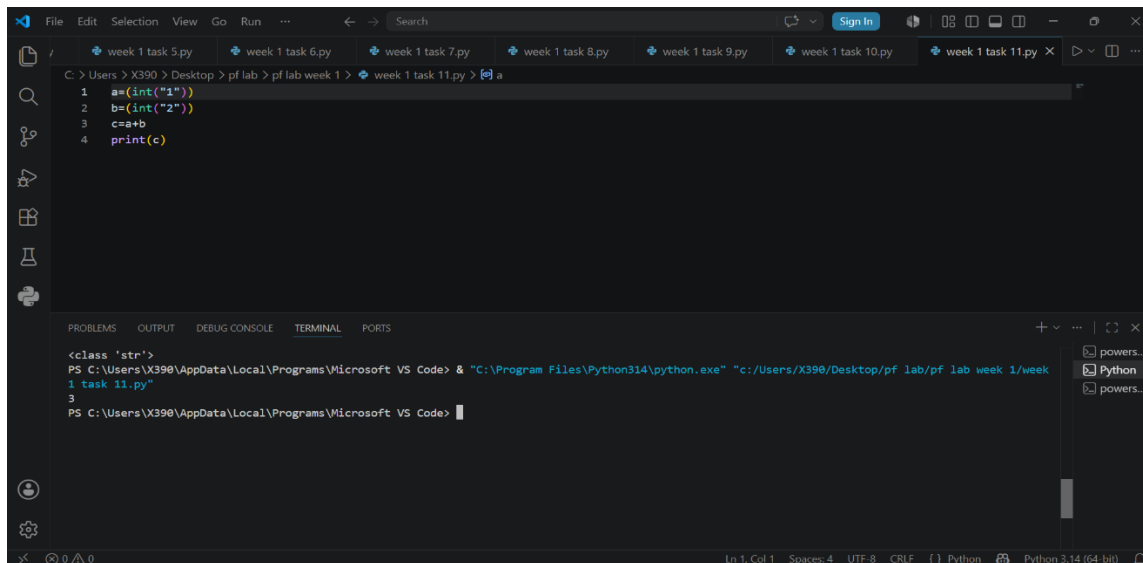
```
File Edit Selection View Go Run ... Search
week 1 task 10.py week 1 task 11.py week 1 task 13.py week 1 task 12.py week 1 task 14.py week 1 task 15.py week 1 task 16.py
C:\Users\X390\Desktop> pf lab > pf lab week 1 > week 1 task 10.py > name
1 name = "zahra"
2 print(name)
3 print(type(name))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
2.4
<class 'float'>
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 1/week
1 task 10.py"
zahra
<class 'str'>
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 1/week
1 task 11.py"
3
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 1/week
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit)
```

TASK NO 11:

Write a Python program to demonstrate explicit data type conversion (type casting) by converting numbers formatted as text strings into actual integers, calculating their sum, and displaying the result.

RESULT:



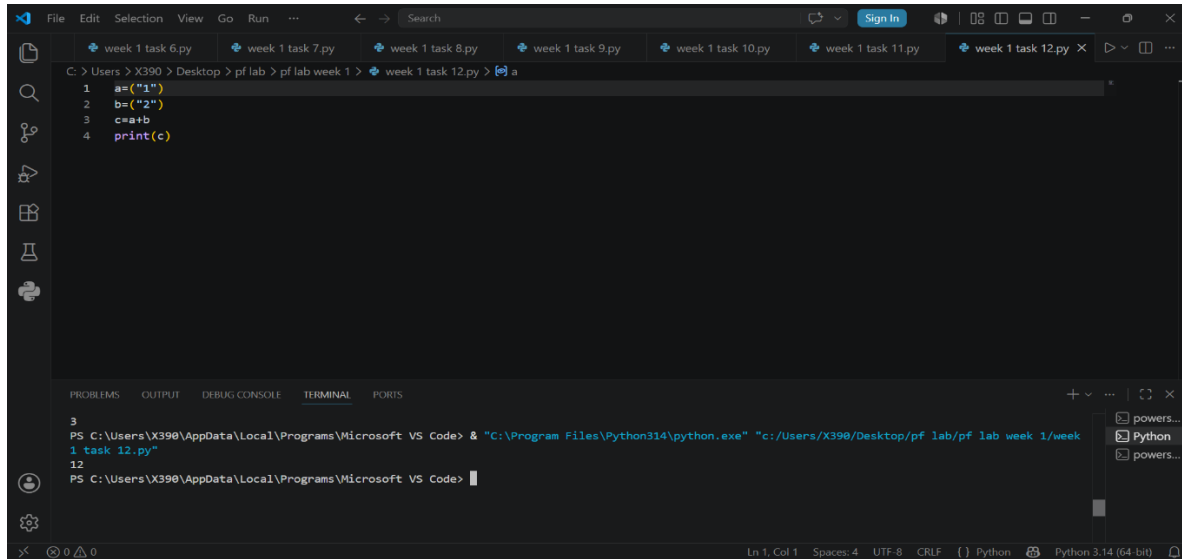
```
File Edit Selection View Go Run ... Search
week 1 task 5.py week 1 task 6.py week 1 task 7.py week 1 task 8.py week 1 task 9.py week 1 task 10.py week 1 task 11.py
C:\Users\X390\Desktop> pf lab > pf lab week 1 > week 1 task 11.py > a
1 a=int("1")
2 b=int("2")
3 c=a+b
4 print(c)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
<class 'str'>
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 1/week
1 task 11.py"
3
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit)
```


TASK NO 12:

Write a Python program to demonstrate how the addition operator (+) behaves when applied to textual data (strings) instead of numerical data types.

RESULT:



The screenshot shows the Visual Studio Code editor with a file named 'week 1 task 12.py'. The code in the editor is as follows:

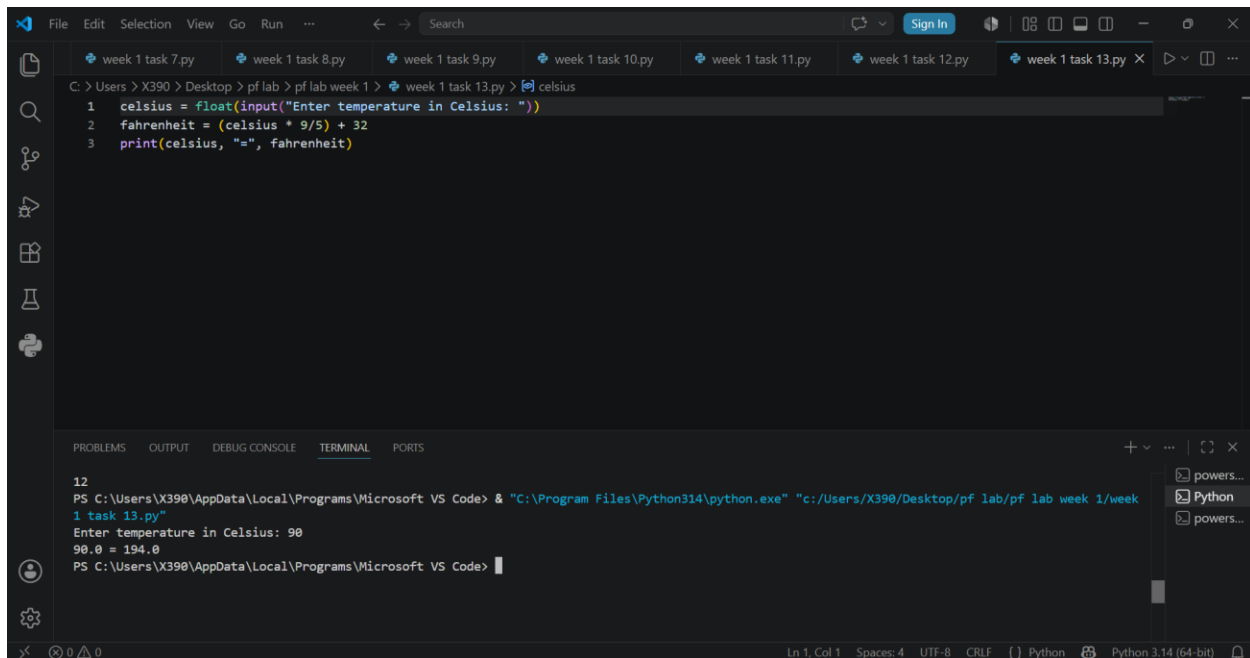
```
1 a="1"
2 b="2"
3 c=a+b
4 print(c)
```

The terminal at the bottom shows the command to run the script: `PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 1/week 1 task 12.py"`. The output of the program is '12', which is the concatenation of the strings '1' and '2'.

TASK NO 13:

Write a Python program that accepts a temperature value in Celsius from a user, programmatically processes it into Fahrenheit using standard mathematical operators, and prints the evaluated result

RESULT:



The screenshot shows the Visual Studio Code editor with a file named 'week 1 task 13.py'. The code in the editor is as follows:

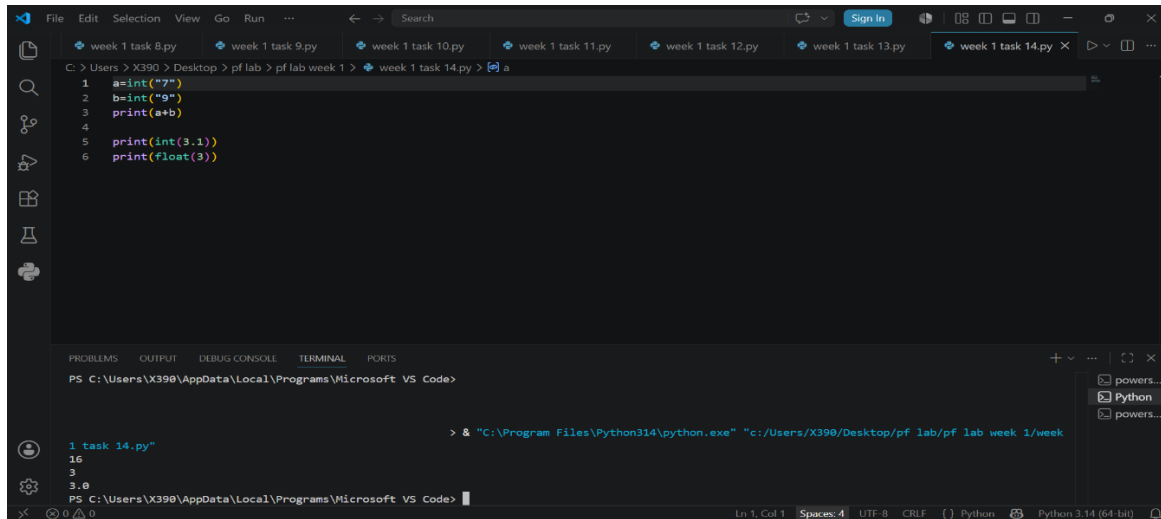
```
1 celsius = float(input("Enter temperature in Celsius: "))
2 fahrenheit = (celsius * 9/5) + 32
3 print(celsius, "=", fahrenheit)
```

The terminal at the bottom shows the command to run the script: `PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 1/week 1 task 13.py"`. The user input is '90', and the output of the program is '90.0 = 194.0', which is the conversion of 90 Celsius to Fahrenheit.

TASK NO 14:

Write a Python program to demonstrate various type casting behaviors using the built-in `int()` and `float()` functions to observe how numeric values change across different data type classifications

RESULT:



```
File Edit Selection View Go Run ... Search
week 1 task 8.py week 1 task 9.py week 1 task 10.py week 1 task 11.py week 1 task 12.py week 1 task 13.py week 1 task 14.py
C:\Users\X390\Desktop> pf lab > pf lab week 1 > week 1 task 14.py > a
1 a=int("7")
2 b=int("9")
3 print(a+b)
4
5 print(int(3.1))
6 print(float(3))

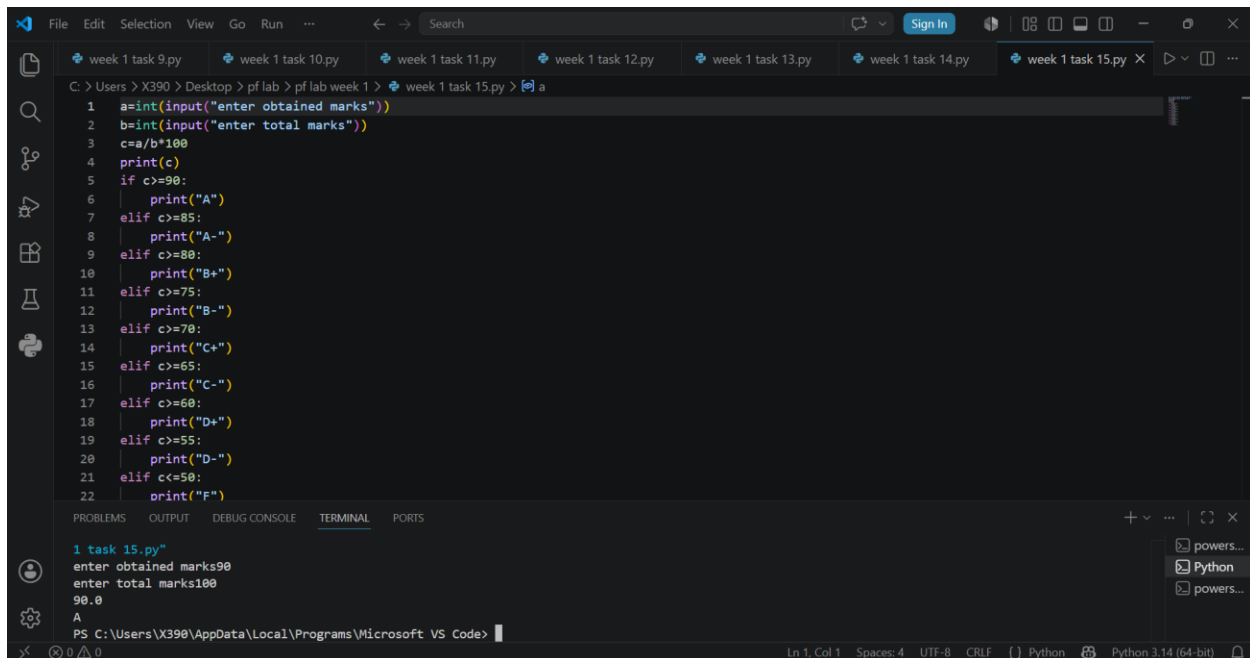
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>

1 task 14.py
16
3
3.0
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>
```

TASK NO 15:

Write a Python program that prompts the user to input a student's obtained marks and total marks, calculates the percentage score, and evaluates a letter grade based on a multi-tiered conditional scale

RESULT:



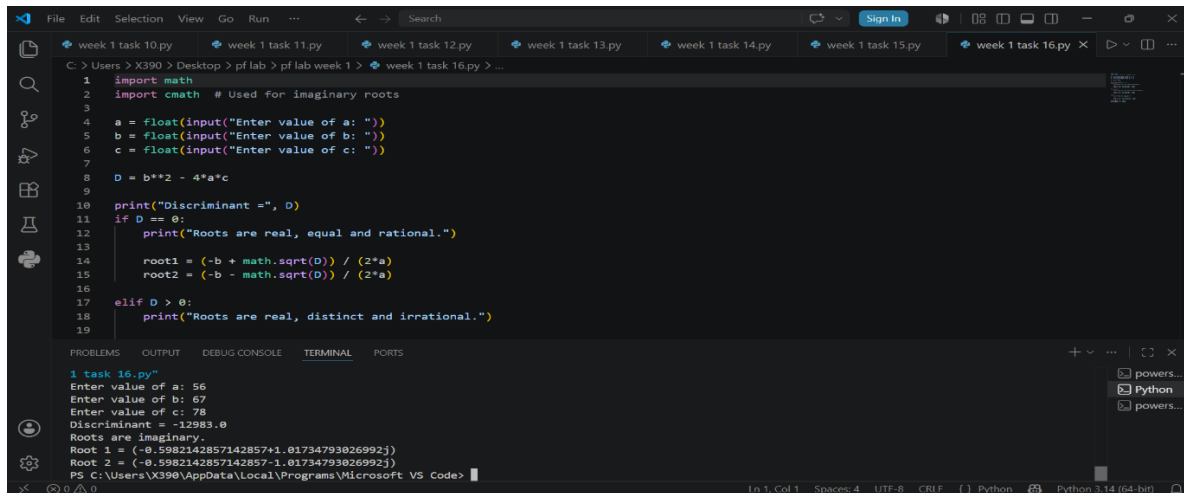
```
File Edit Selection View Go Run ... Search
week 1 task 9.py week 1 task 10.py week 1 task 11.py week 1 task 12.py week 1 task 13.py week 1 task 14.py week 1 task 15.py
C:\Users\X390\Desktop> pf lab > pf lab week 1 > week 1 task 15.py > a
1 a=int(input("enter obtained marks"))
2 b=int(input("enter total marks"))
3 c=a/b*100
4 print(c)
5 if c>=90:
6     print("A")
7 elif c>=85:
8     print("A-")
9 elif c>=80:
10    print("B+")
11 elif c>=75:
12    print("B-")
13 elif c>=70:
14    print("C+")
15 elif c>=65:
16    print("C-")
17 elif c>=60:
18    print("D+")
19 elif c>=55:
20    print("D-")
21 elif c<=50:
22    print("F")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
1 task 15.py
enter obtained marks90
enter total marks100
90.0
A
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>
```

TASK NO 16:

Write a Python program that calculates the discriminant and finds the roots of a quadratic equation $ax^2 + bx + c = 0$ given the coefficients a , b , and c . The program must identify the nature of the roots (real and equal, real and distinct, or imaginary) and calculate them using appropriate mathematical modules.

RESULT:



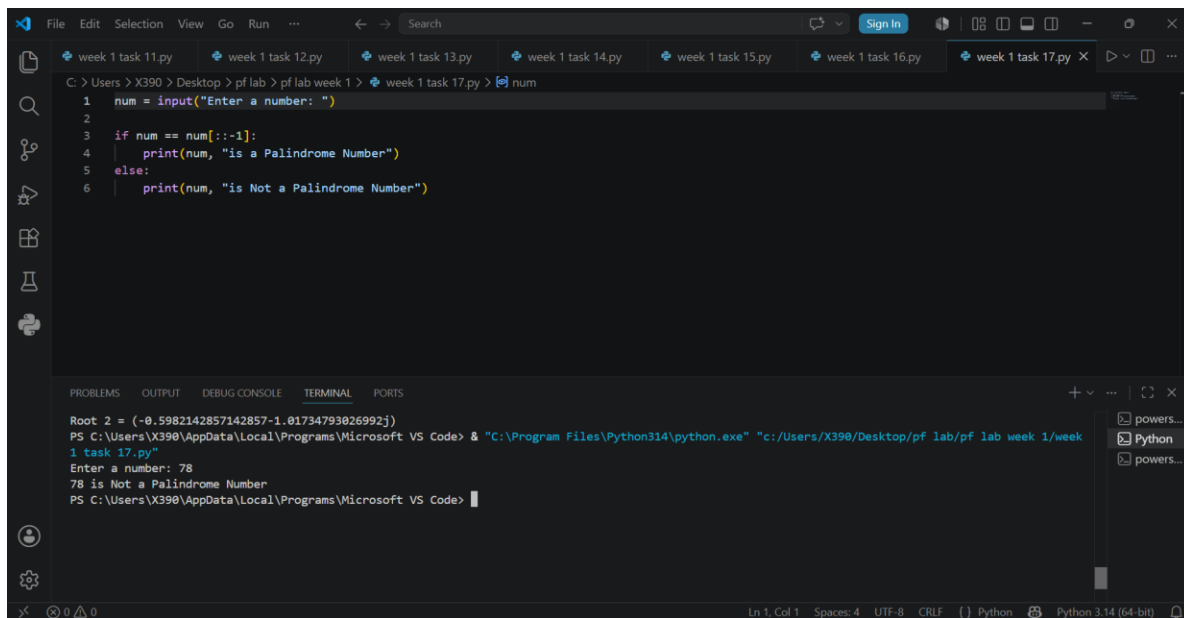
```
1 import math
2 import cmath # Used for imaginary roots
3
4 a = float(input("Enter value of a: "))
5 b = float(input("Enter value of b: "))
6 c = float(input("Enter value of c: "))
7
8 D = b**2 - 4*a*c
9
10 print("Discriminant =", D)
11 if D == 0:
12     print("Roots are real, equal and rational.")
13
14     root1 = (-b + math.sqrt(D)) / (2*a)
15     root2 = (-b - math.sqrt(D)) / (2*a)
16
17 elif D > 0:
18     print("Roots are real, distinct and irrational.")
19
20
21
22
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30
31
32
33
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97
98
99
100
```

1 task 16.py
Enter value of a: 56
Enter value of b: 67
Enter value of c: 78
Discriminant = -12983.0
Roots are imaginary.
Root 1 = (-0.5982142857142857+1.01734793026992j)
Root 2 = (-0.5982142857142857-1.01734793026992j)
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>

TASK NO 17:

Write a Python program that accepts a sequence of characters or digits from the user and checks whether it qualifies as a palindrome

RESULT:



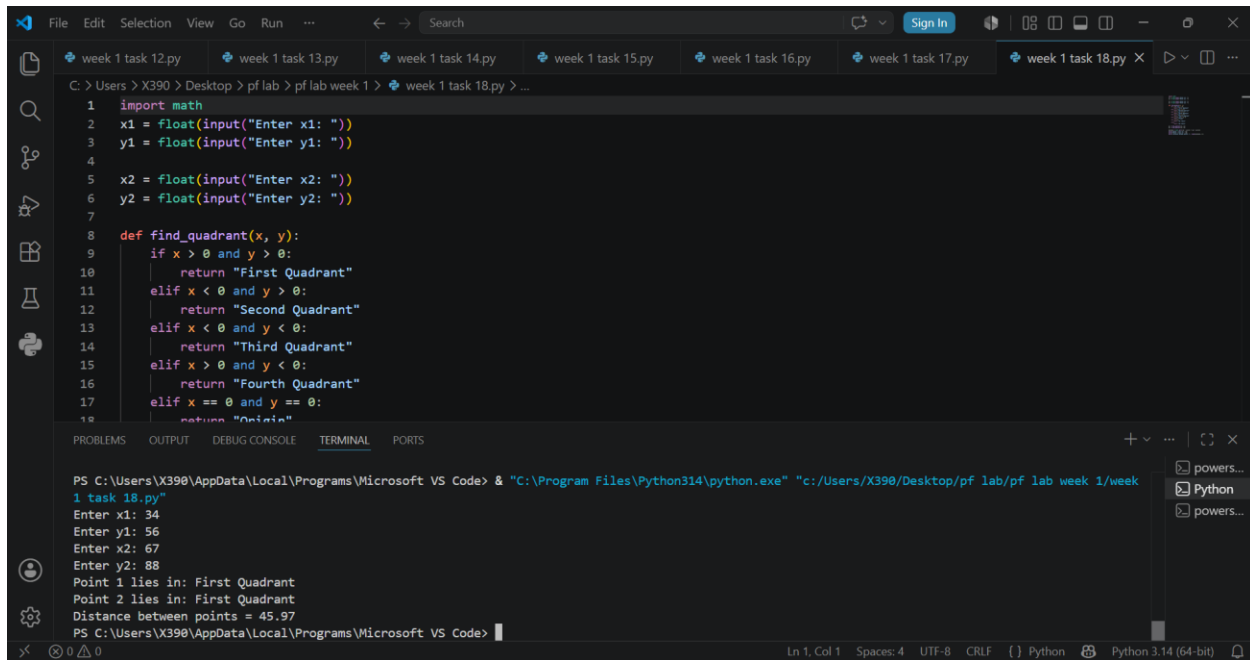
```
1 num = input("Enter a number: ")
2
3 if num == num[::-1]:
4     print(num, "is a Palindrome Number")
5 else:
6     print(num, "is Not a Palindrome Number")
7
8
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91
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93
94
95
96
97
98
99
100
```

1 task 17.py
Enter a number: 78
78 is Not a Palindrome Number
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>

TASK NO 18:

Write a Python program that accepts the \$(x, y)\$ coordinates for two distinct points on a 2D Cartesian plane, determines which quadrant each individual point resides in, and calculates the Euclidean geometric distance between them.

RESULT:



The screenshot shows a VS Code editor with a Python file named 'week 1 task 18.py'. The code defines a function 'find_quadrant(x, y)' that returns the quadrant of a point (x, y) based on its coordinates. It also calculates the Euclidean distance between two points (x1, y1) and (x2, y2). The terminal output shows the program running and displaying the results for the input points (34, 56) and (67, 88).

```
1 import math
2 x1 = float(input("Enter x1: "))
3 y1 = float(input("Enter y1: "))
4
5 x2 = float(input("Enter x2: "))
6 y2 = float(input("Enter y2: "))
7
8 def find_quadrant(x, y):
9     if x > 0 and y > 0:
10         return "First Quadrant"
11     elif x < 0 and y > 0:
12         return "Second Quadrant"
13     elif x < 0 and y < 0:
14         return "Third Quadrant"
15     elif x > 0 and y < 0:
16         return "Fourth Quadrant"
17     elif x == 0 and y == 0:
18         return "Origin"
19
20 distance = math.sqrt((x2 - x1)**2 + (y2 - y1)**2)
21 print("Distance between points = ", distance)
```

Terminal Output:

```
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 1/week 1 task 18.py"
Enter x1: 34
Enter y1: 56
Enter x2: 67
Enter y2: 88
Point 1 lies in: First Quadrant
Point 2 lies in: First Quadrant
Distance between points = 45.97
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>
```

CONCLUSION:

In conclusion, mastering input/output mechanisms and core data types allows programs to interact with users and manage diverse informational structures efficiently. Operators act as the functional engine, driving mathematical calculations, value comparisons, and logical evaluations across these data types. Conditional statements (\$if/else\$) complete this fundamental framework by introducing decision-making logic, directing the program's execution path based on dynamic runtime data. Together, these core programming concepts transform static code into interactive, logical, and adaptive software applications. Understanding these principles is essential, as they form the foundational building blocks for all complex algorithms and advanced system architectures